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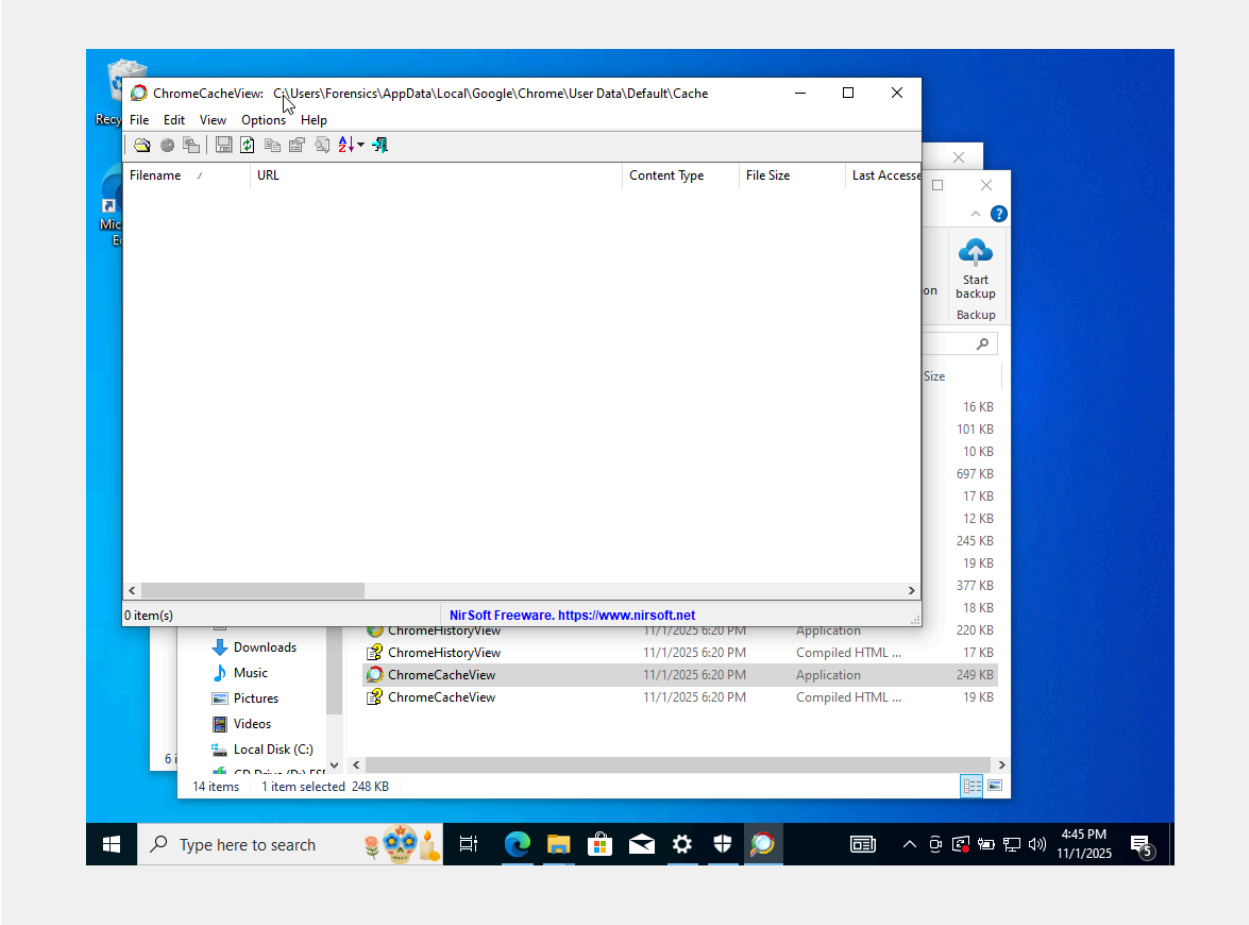
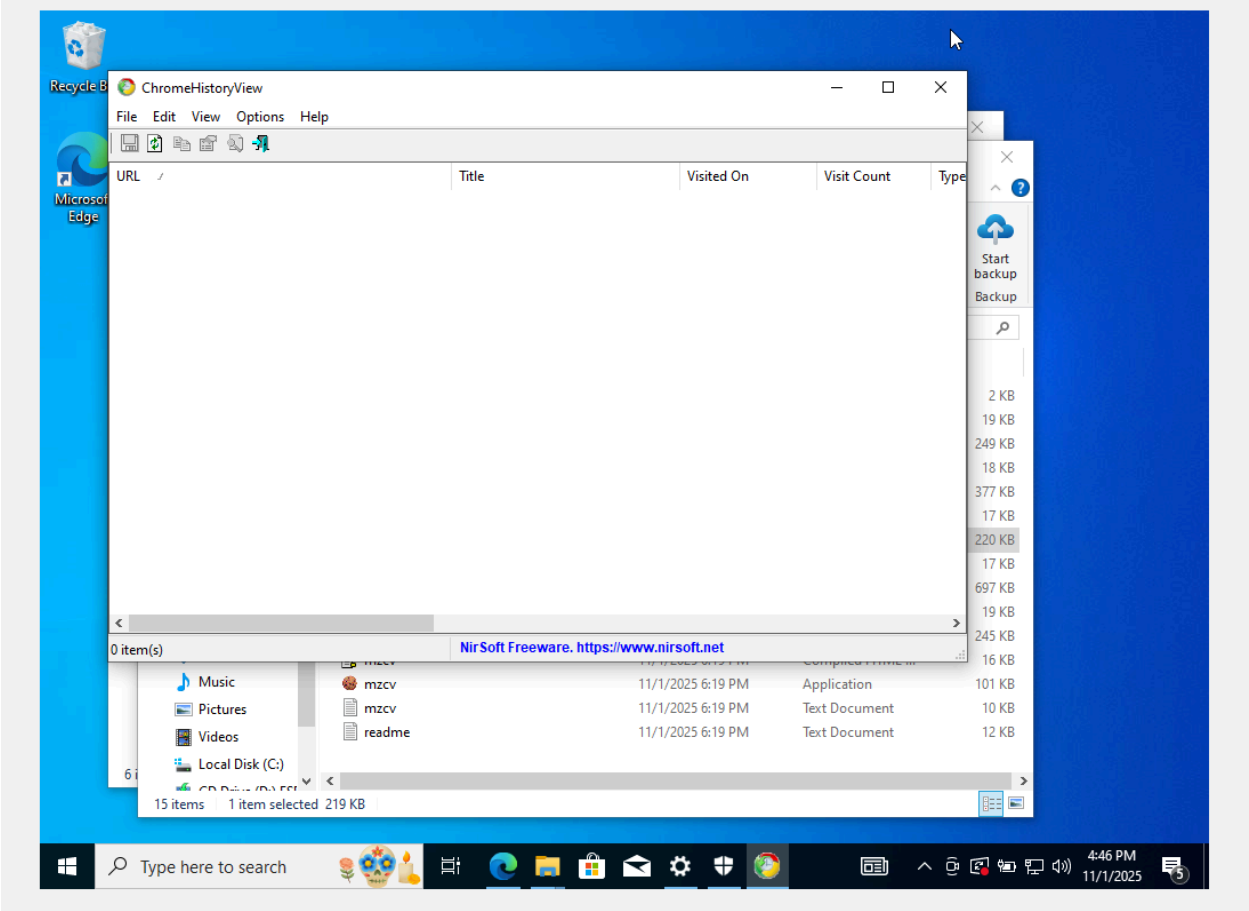
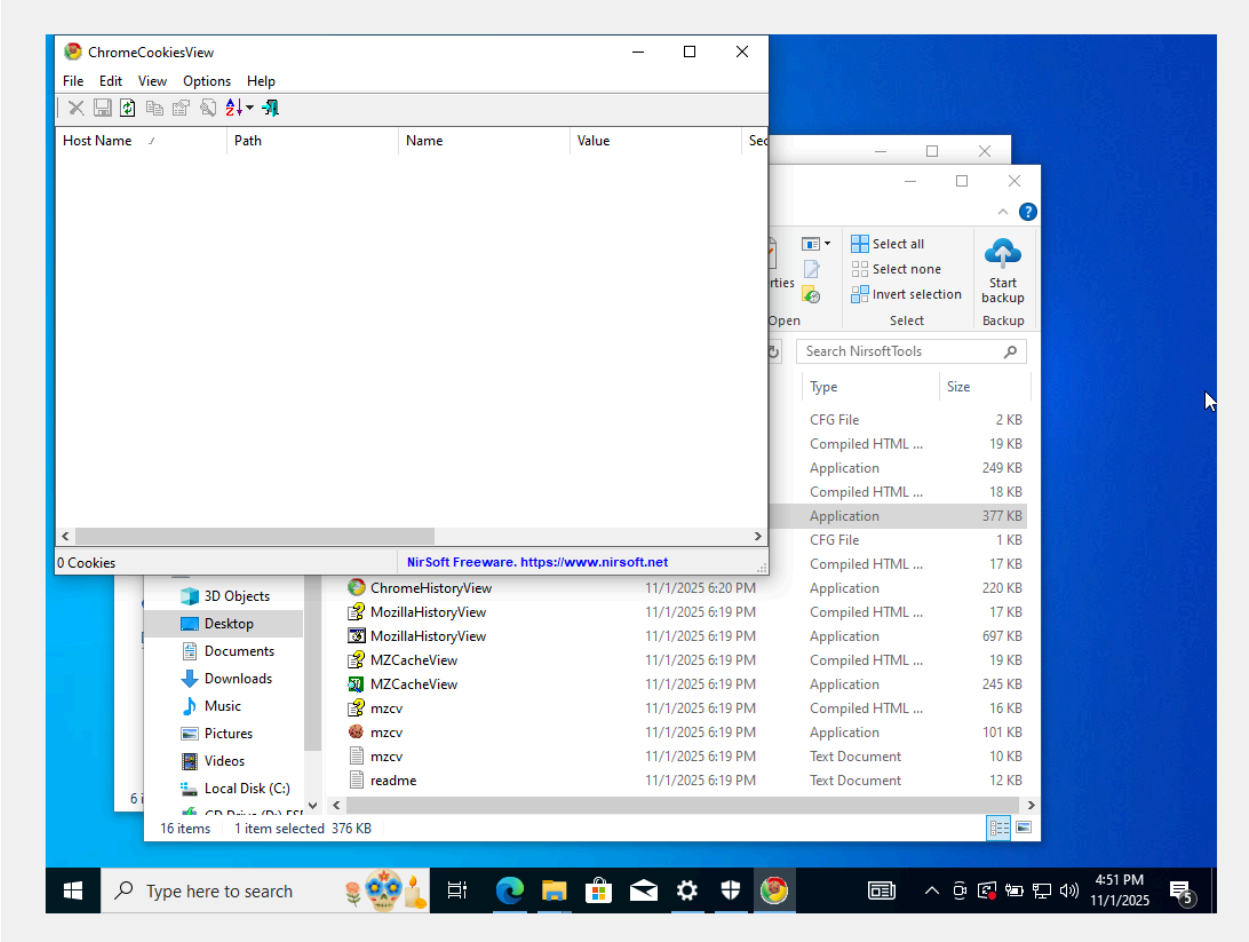
Windows User Activity Investigation

In this lab, I explored different Windows forensic tools and techniques to analyze browser data, event logs, registry information, and other important system artifacts. The goal was to understand how Windows stores and tracks user activity through built-in and third-party tools. I used NirSoft utilities, Event Log Explorer, and Python scripts to extract and analyze digital evidence such as cache files, user accounts, registry timestamps, and browser history.

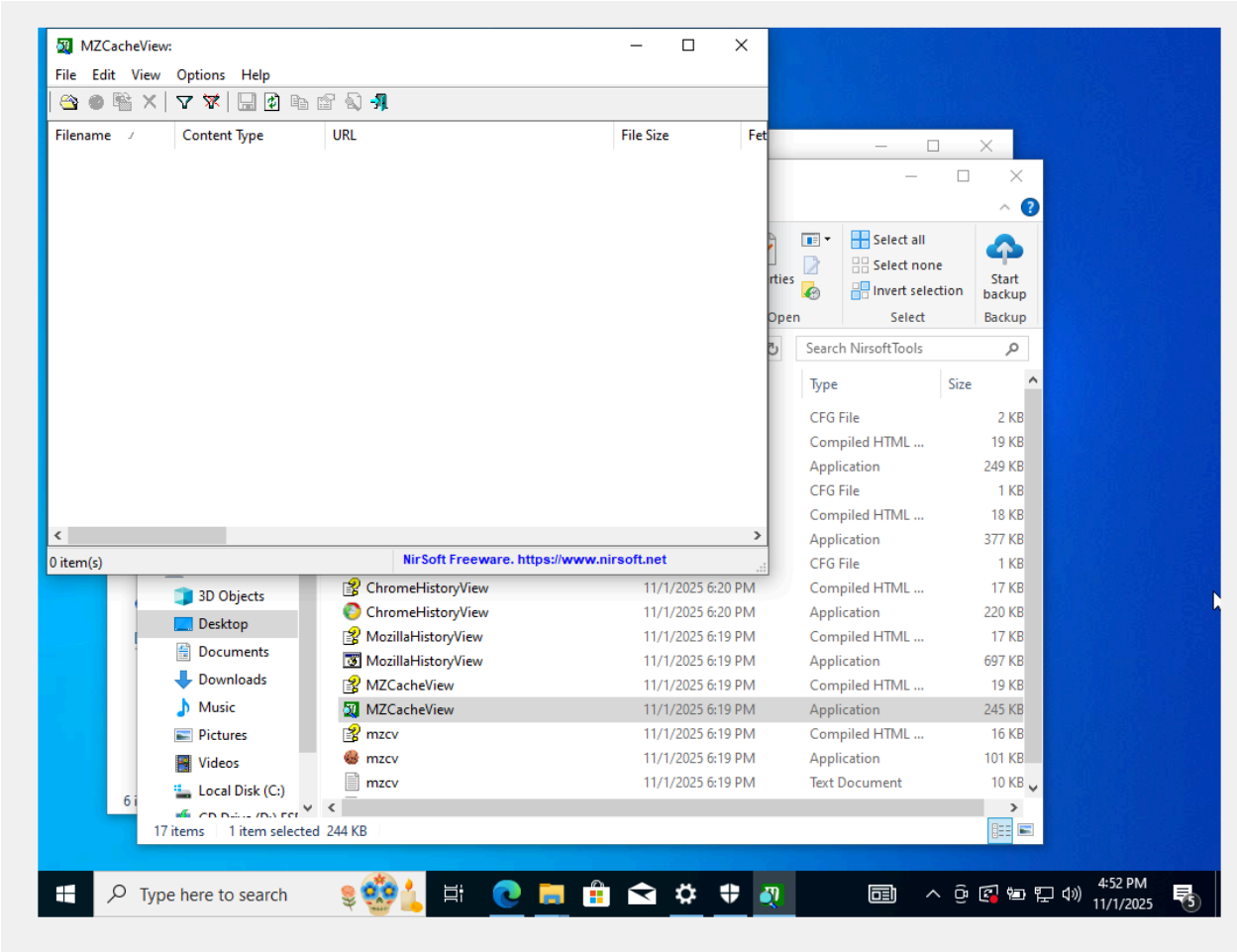
I began by analyzing browser data using NirSoft's utilities. For Chrome, I used ChromeCacheView, ChromeHistoryView, and ChromeCookiesView to examine the user's cache, browsing history, and cookies stored in the local profile directory.

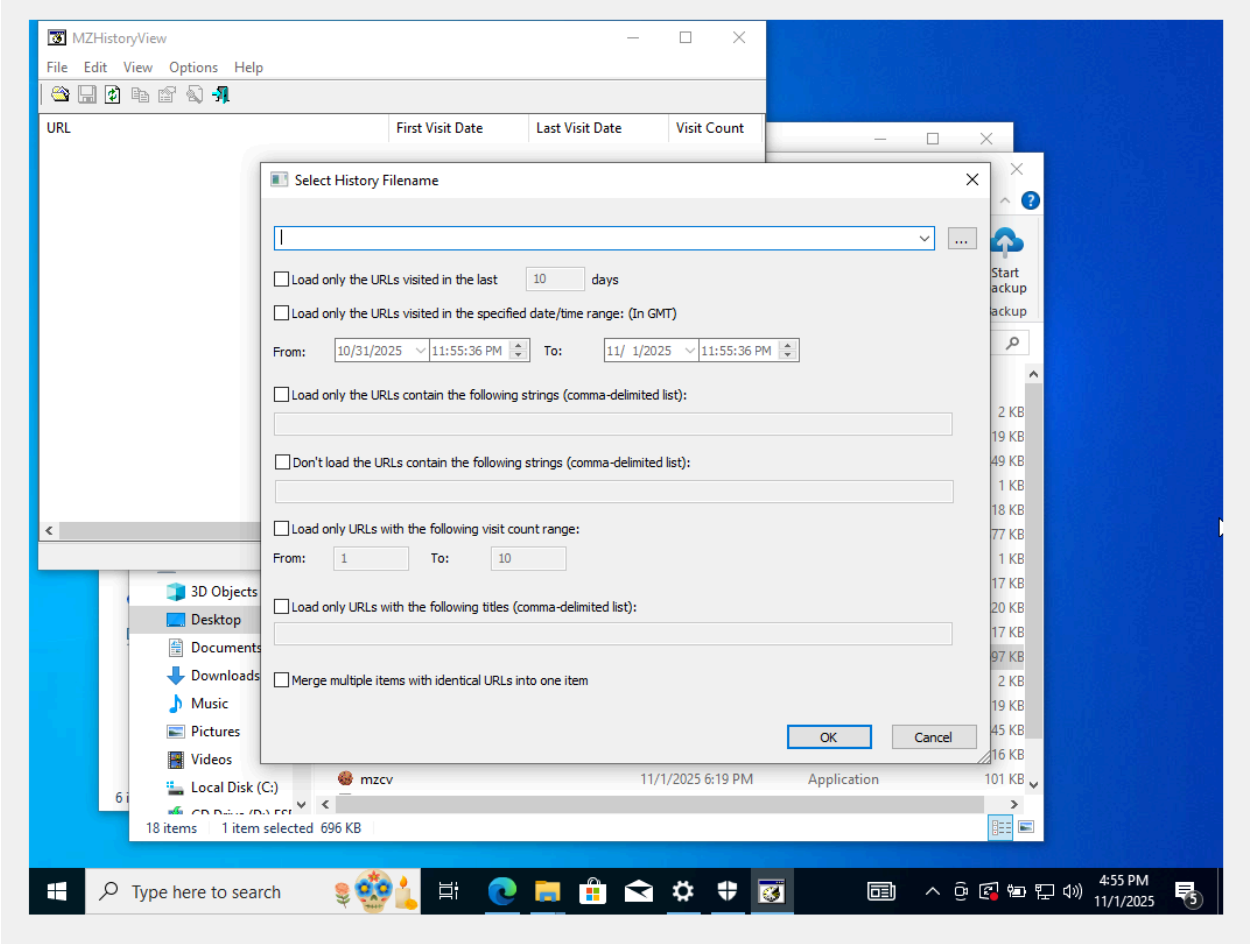


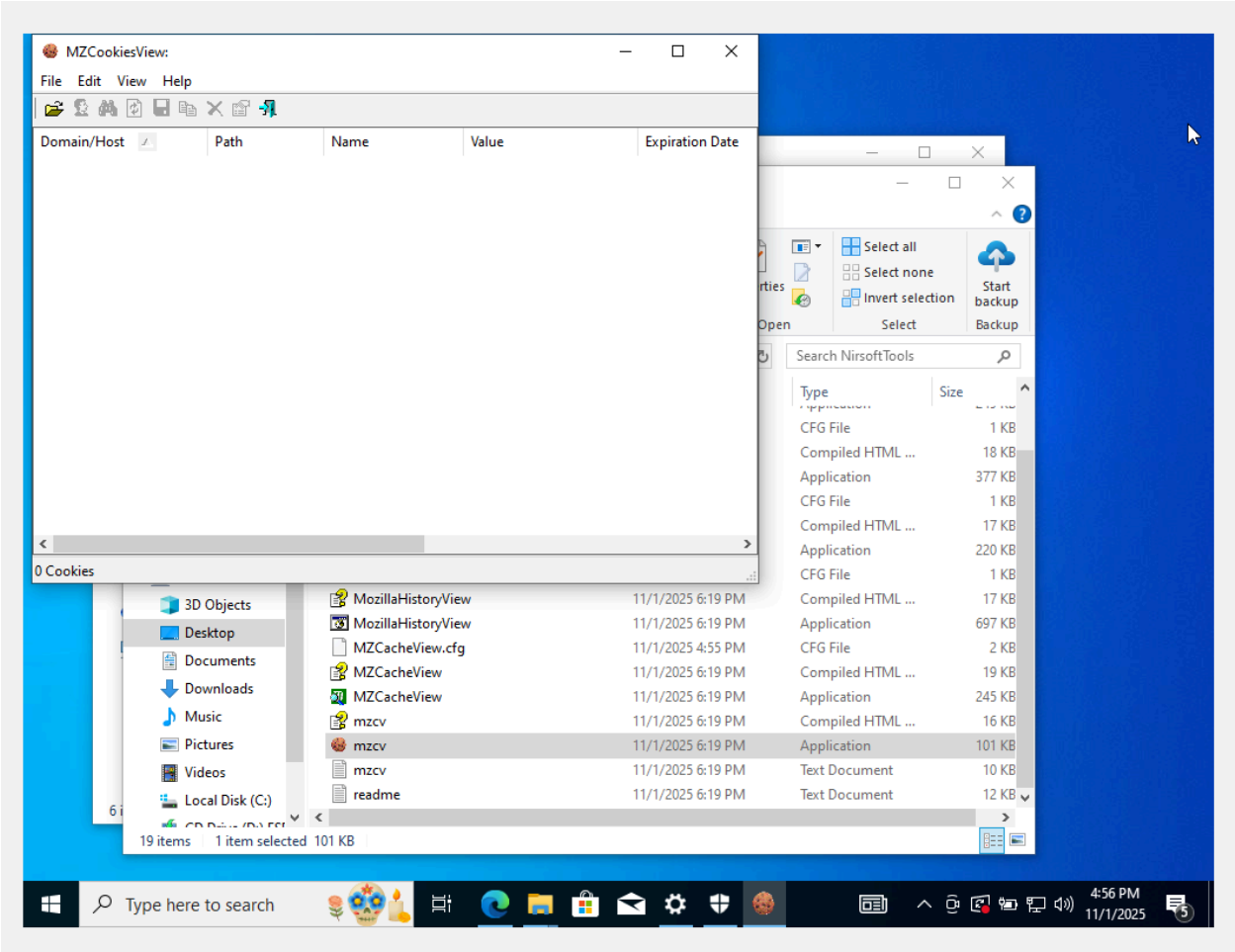

ChromeHistoryView
File Edit View Options Help
URL
Title
Visited On
Visit Count
Type
0 item(s)
NirSoft Freeware. https://www.nirsoft.net
Music
Pictures
Videos
Local Disk (C:)
mzcv 11/1/2025 6:19 PM Application 101 KB
mzcv 11/1/2025 6:19 PM Text Document 10 KB
readme 11/1/2025 6:19 PM Text Document 12 KB
15 items 1 item selected 219 KB
Start backup
Backup
Type here to search
4:46 PM
11/1/2025




Next, I used similar tools for Mozilla Firefox. These included MZCacheView, MZHistoryView, and MZCookiesView. These tools allowed me to look at how Firefox stores cache, cookies, and visited URLs.


MZCacheView:
File Edit View Options Help
Filename
Content Type
URL
File Size
0 item(s)
NirSoft Freeware. https://www.nirsoft.net
Select all
Select none
Invert selection
Select
Start backup
Backup
Open
Search NirsoftTools
Type
Size
CFG File 2 KB
Compiled HTML ... 19 KB
Application 249 KB
CFG File 1 KB
Compiled HTML ... 18 KB
Application 377 KB
CFG File 1 KB
Compiled HTML ... 17 KB
Application 220 KB
Compiled HTML ... 17 KB
Application 697 KB
Compiled HTML ... 19 KB
Application 245 KB
Compiled HTML ... 16 KB
Application 101 KB
Text Document 10 KB
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
ChromeHistoryView 11/1/2025 6:20 PM
ChromeHistoryView 11/1/2025 6:20 PM
MozillaHistoryView 11/1/2025 6:19 PM
MozillaHistoryView 11/1/2025 6:19 PM
MZCacheView 11/1/2025 6:19 PM
MZCacheView 11/1/2025 6:19 PM
mzcv 11/1/2025 6:19 PM
mzcv 11/1/2025 6:19 PM
mzcv 11/1/2025 6:19 PM
17 items
1 item selected 244 KB
Type here to search
4:52 PM
11/1/2025

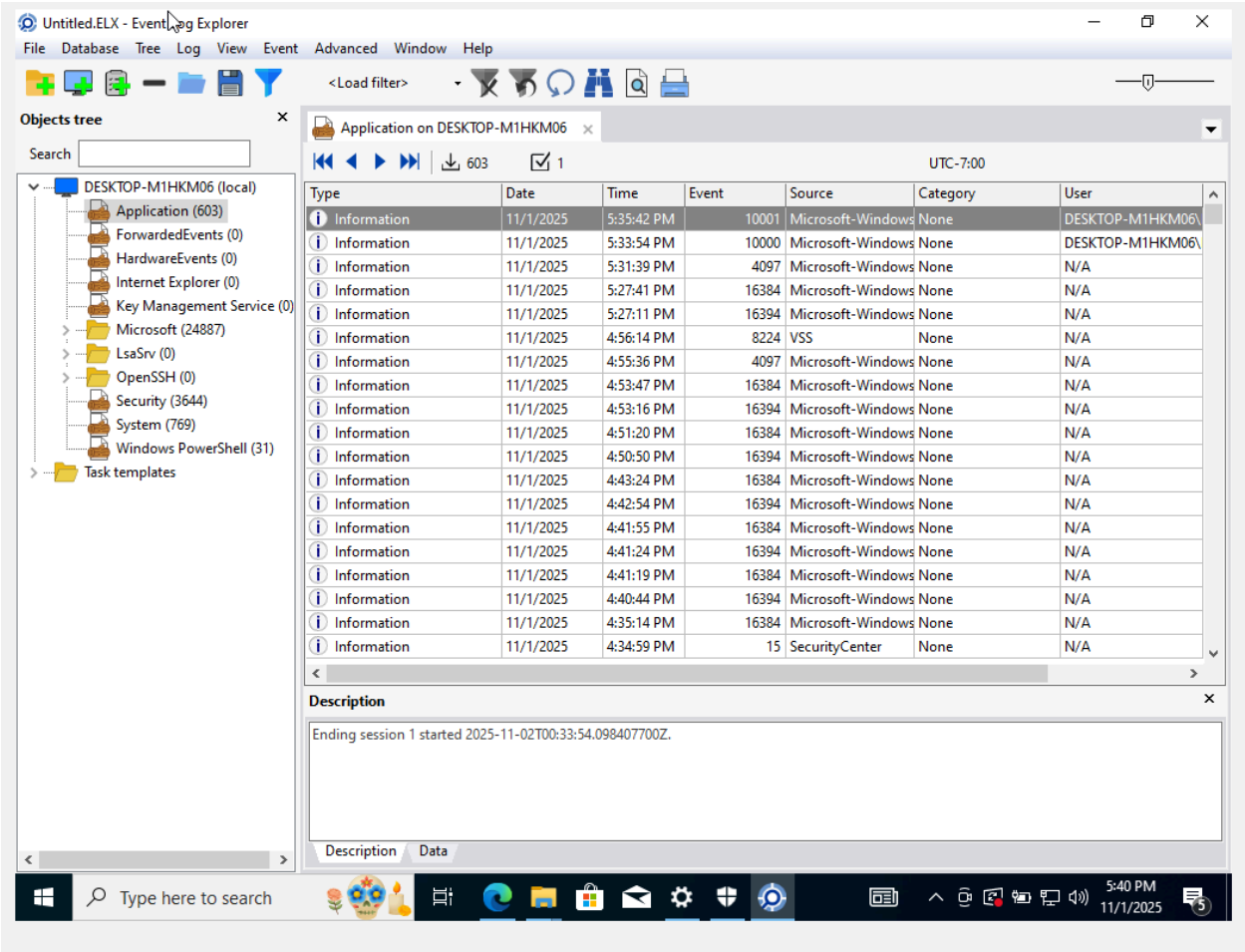

MZHistoryView
File Edit View Options Help
URL
First Visit Date
Last Visit Date
Visit Count
Select History Filename
Load only the URLs visited in the last 10 days
Load only the URLs visited in the specified date/time range: (In GMT)
From: 10/31/2025 11:55:36 PM To: 11/ 1/2025 11:55:36 PM
Load only the URLs contain the following strings (comma-delimited list):
Don't load the URLs contain the following strings (comma-delimited list):
Load only URLs with the following visit count range:
From: 1 To: 10
Load only URLs with the following titles (comma-delimited list):
Merge multiple items with identical URLs into one item
OK
Cancel
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
mzcv
11/1/2025 6:19 PM
Application
101 KB
18 items
1 item selected 696 KB
Type here to search
4:55 PM
11/1/2025

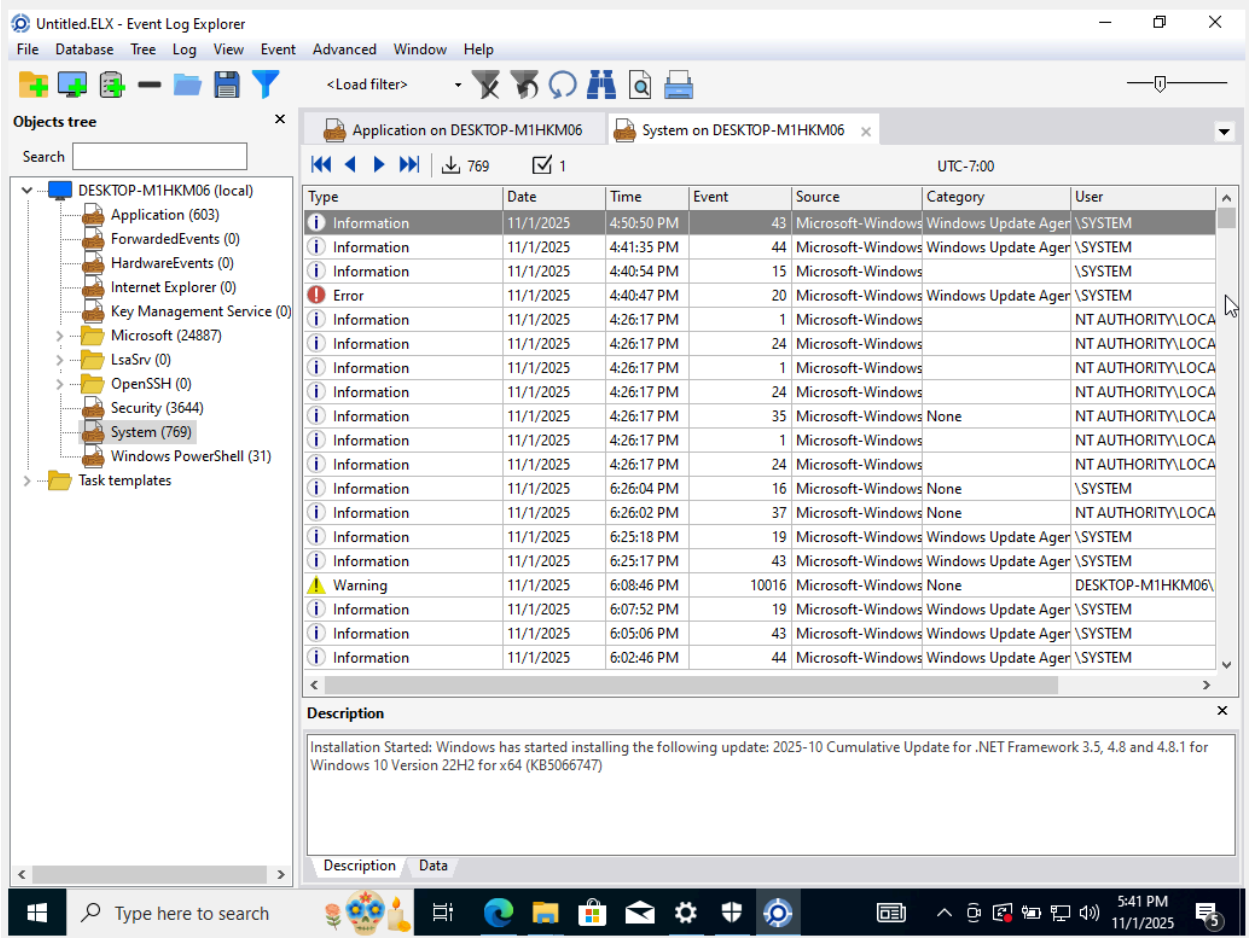



Even when no active data appeared, these screenshots show that I successfully located and accessed the correct forensic paths for both Chrome and Firefox data.

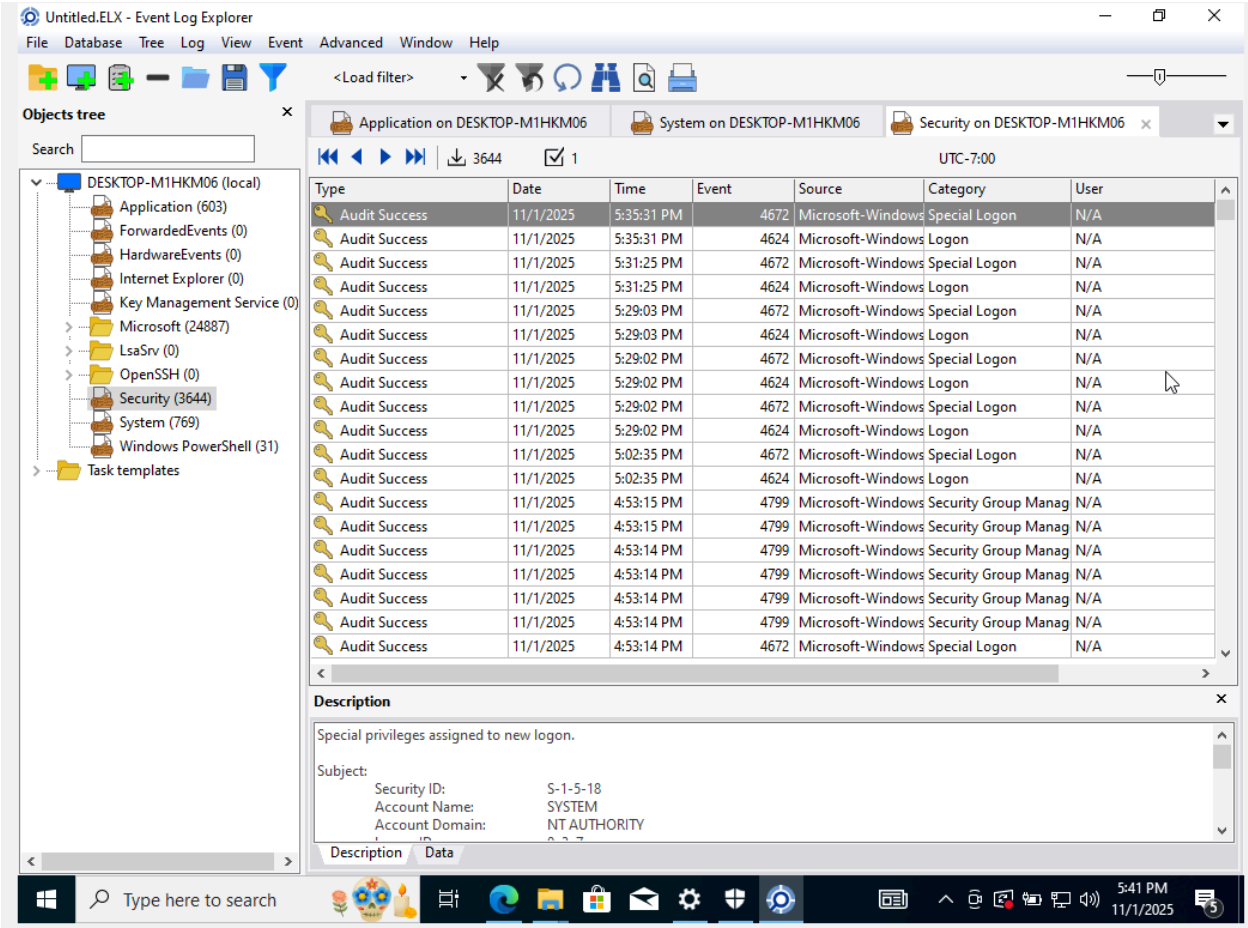
Part II – Windows Event Log Analysis

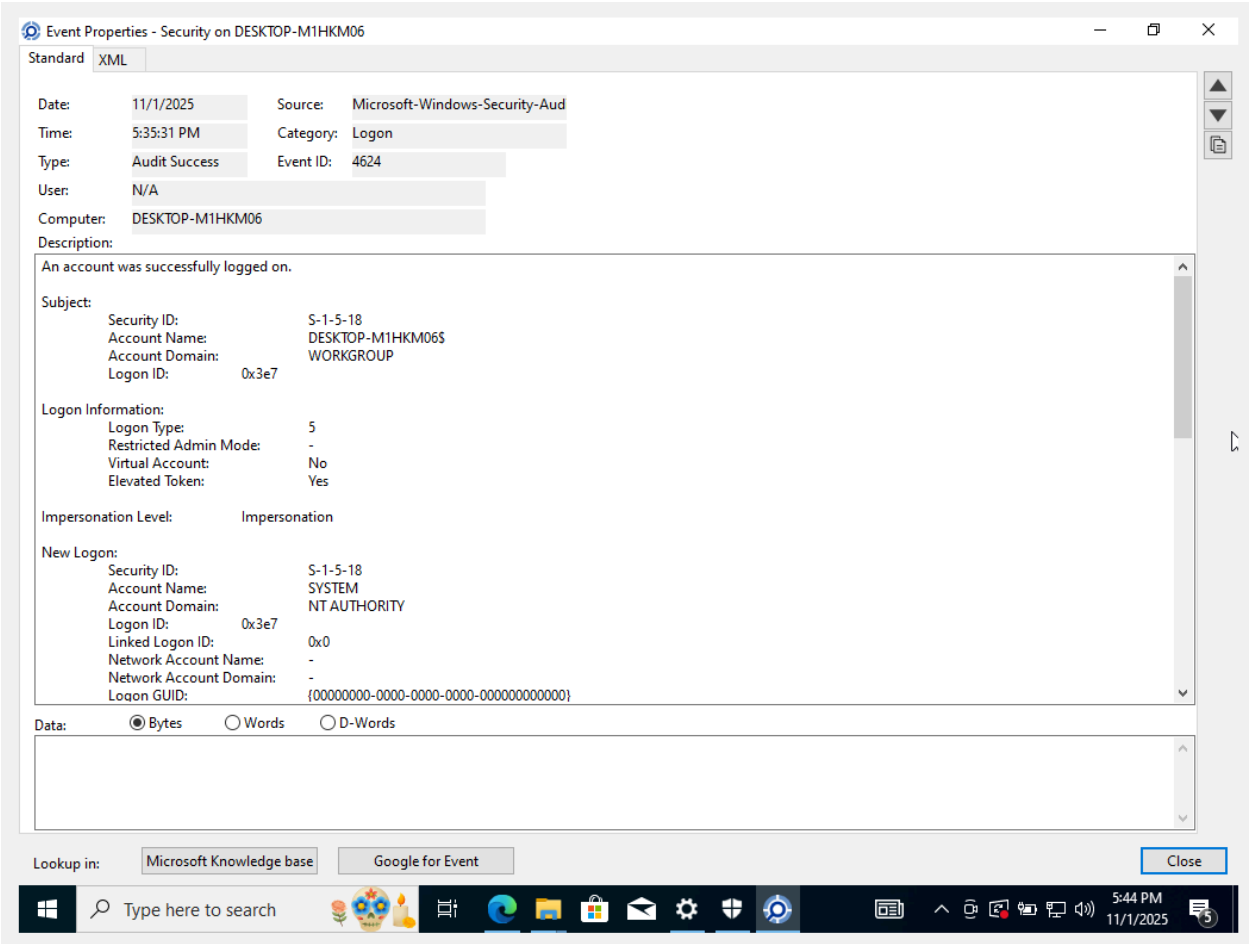
For this section, I used Event Log Explorer to analyze Application, System, and Security logs. These logs record system activities, user logons, software installations, and Windows updates.


Untitled.ELX - Event Log Explorer
File Database Tree Log View Event Advanced Window Help
<Load filter>
Objects tree
Search
DESKTOP-M1HKM06 (local)
Application (603)
ForwardedEvents (0)
HardwareEvents (0)
Internet Explorer (0)
Key Management Service (0)
Microsoft (24887)
LsaSrv (0)
OpenSSH (0)
Security (3644)
System (769)
Windows PowerShell (31)
Task templates
Application on DESKTOP-M1HKM06
603
1
UTC-7:00
Type Date Time Event Source Category User
Information 11/1/2025 5:35:42 PM 10001 Microsoft-Windows None DESKTOP-M1HKM06\
Information 11/1/2025 5:33:54 PM 10000 Microsoft-Windows None DESKTOP-M1HKM06\
Information 11/1/2025 5:31:39 PM 4097 Microsoft-Windows None N/A
Information 11/1/2025 5:27:41 PM 16384 Microsoft-Windows None N/A
Information 11/1/2025 5:27:11 PM 16394 Microsoft-Windows None N/A
Information 11/1/2025 4:56:14 PM 8224 VSS None N/A
Information 11/1/2025 4:55:36 PM 4097 Microsoft-Windows None N/A
Information 11/1/2025 4:53:47 PM 16384 Microsoft-Windows None N/A
Information 11/1/2025 4:53:16 PM 16394 Microsoft-Windows None N/A
Information 11/1/2025 4:51:20 PM 16384 Microsoft-Windows None N/A
Information 11/1/2025 4:50:50 PM 16394 Microsoft-Windows None N/A
Information 11/1/2025 4:43:24 PM 16384 Microsoft-Windows None N/A
Information 11/1/2025 4:42:54 PM 16394 Microsoft-Windows None N/A
Information 11/1/2025 4:41:55 PM 16384 Microsoft-Windows None N/A
Information 11/1/2025 4:41:24 PM 16394 Microsoft-Windows None N/A
Information 11/1/2025 4:41:19 PM 16384 Microsoft-Windows None N/A
Information 11/1/2025 4:40:44 PM 16394 Microsoft-Windows None N/A
Information 11/1/2025 4:35:14 PM 16384 Microsoft-Windows None N/A
Information 11/1/2025 4:34:59 PM 15 SecurityCenter None N/A
Description
Ending session 1 started 2025-11-02T00:33:54.098407700Z.
Description Data
Type here to search
5:40 PM
11/1/2025

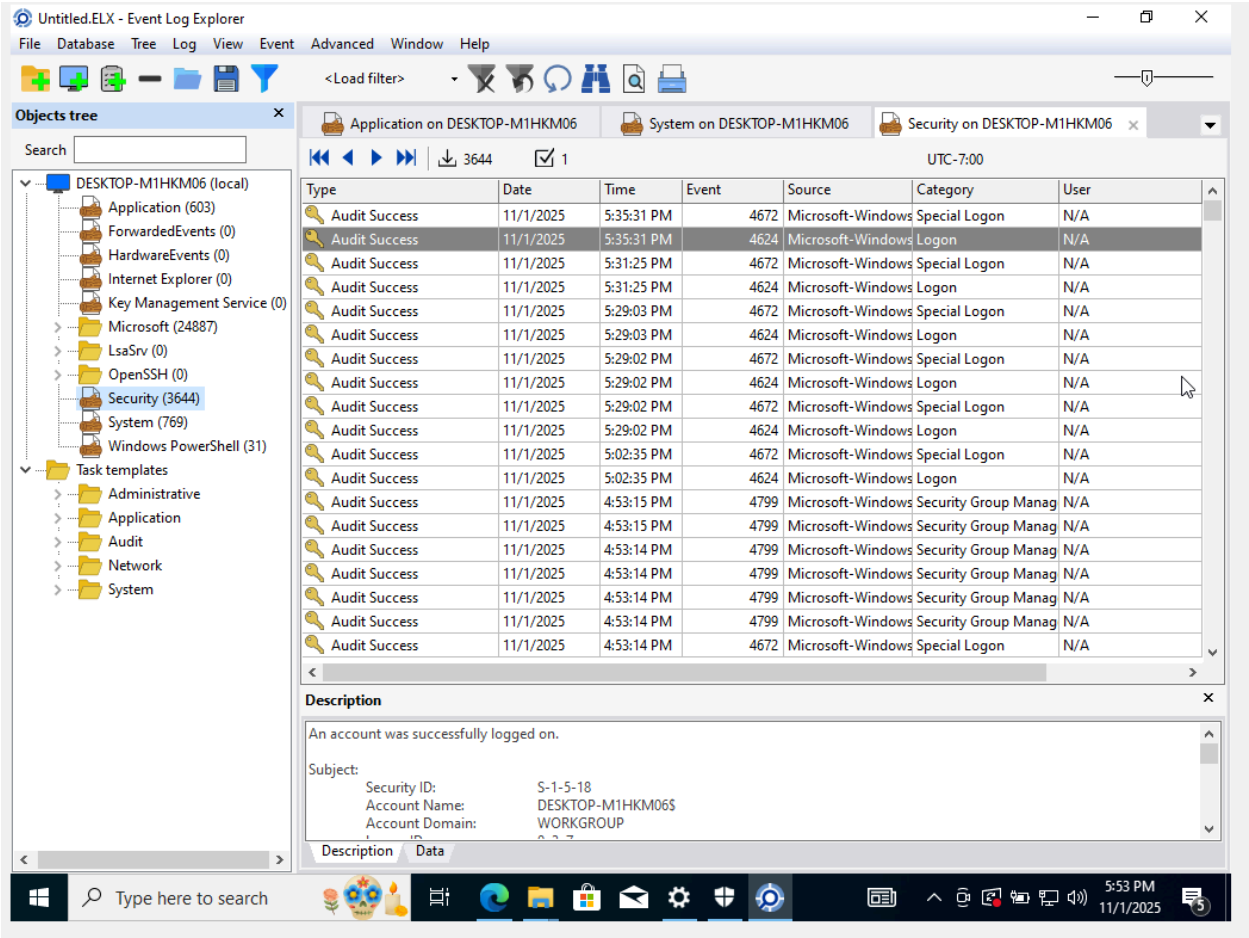



Then, I viewed Security logs that track user login and logoff events.


Untitled.ELX - Event Log Explorer
File Database Tree Log View Event Advanced Window Help
<Load filter>
Objects tree
Search
DESKTOP-M1HKM06 (local)
Application (603)
ForwardedEvents (0)
HardwareEvents (0)
Internet Explorer (0)
Key Management Service (0)
Microsoft (24887)
LsaSrv (0)
OpenSSH (0)
Security (3644)
System (769)
Windows PowerShell (31)
Task templates
Application on DESKTOP-M1HKM06
System on DESKTOP-M1HKM06
Security on DESKTOP-M1HKM06
3644
1
UTC-7:00
Type Date Time Event Source Category User
Audit Success 11/1/2025 5:35:31 PM 4672 Microsoft-Windows Special Logon N/A
Audit Success 11/1/2025 5:35:31 PM 4624 Microsoft-Windows Logon N/A
Audit Success 11/1/2025 5:31:25 PM 4672 Microsoft-Windows Special Logon N/A
Audit Success 11/1/2025 5:31:25 PM 4624 Microsoft-Windows Logon N/A
Audit Success 11/1/2025 5:29:03 PM 4672 Microsoft-Windows Special Logon N/A
Audit Success 11/1/2025 5:29:03 PM 4624 Microsoft-Windows Logon N/A
Audit Success 11/1/2025 5:29:02 PM 4672 Microsoft-Windows Special Logon N/A
Audit Success 11/1/2025 5:29:02 PM 4624 Microsoft-Windows Logon N/A
Audit Success 11/1/2025 5:29:02 PM 4672 Microsoft-Windows Special Logon N/A
Audit Success 11/1/2025 5:29:02 PM 4624 Microsoft-Windows Logon N/A
Audit Success 11/1/2025 5:02:35 PM 4672 Microsoft-Windows Special Logon N/A
Audit Success 11/1/2025 5:02:35 PM 4624 Microsoft-Windows Logon N/A
Audit Success 11/1/2025 4:53:15 PM 4799 Microsoft-Windows Security Group Manag N/A
Audit Success 11/1/2025 4:53:15 PM 4799 Microsoft-Windows Security Group Manag N/A
Audit Success 11/1/2025 4:53:14 PM 4799 Microsoft-Windows Security Group Manag N/A
Audit Success 11/1/2025 4:53:14 PM 4799 Microsoft-Windows Security Group Manag N/A
Audit Success 11/1/2025 4:53:14 PM 4799 Microsoft-Windows Security Group Manag N/A
Audit Success 11/1/2025 4:53:14 PM 4799 Microsoft-Windows Security Group Manag N/A
Audit Success 11/1/2025 4:53:14 PM 4672 Microsoft-Windows Special Logon N/A
Description
Special privileges assigned to new logon.
Subject:
Security ID: S-1-5-18
Account Name: SYSTEM
Account Domain: NT AUTHORITY
Description Data
Type here to search
5:41 PM
11/1/2025




Finally, I applied a filter under the “Task Templates” section to focus on Audit Success logon events, which help identify user access times and security privileges.



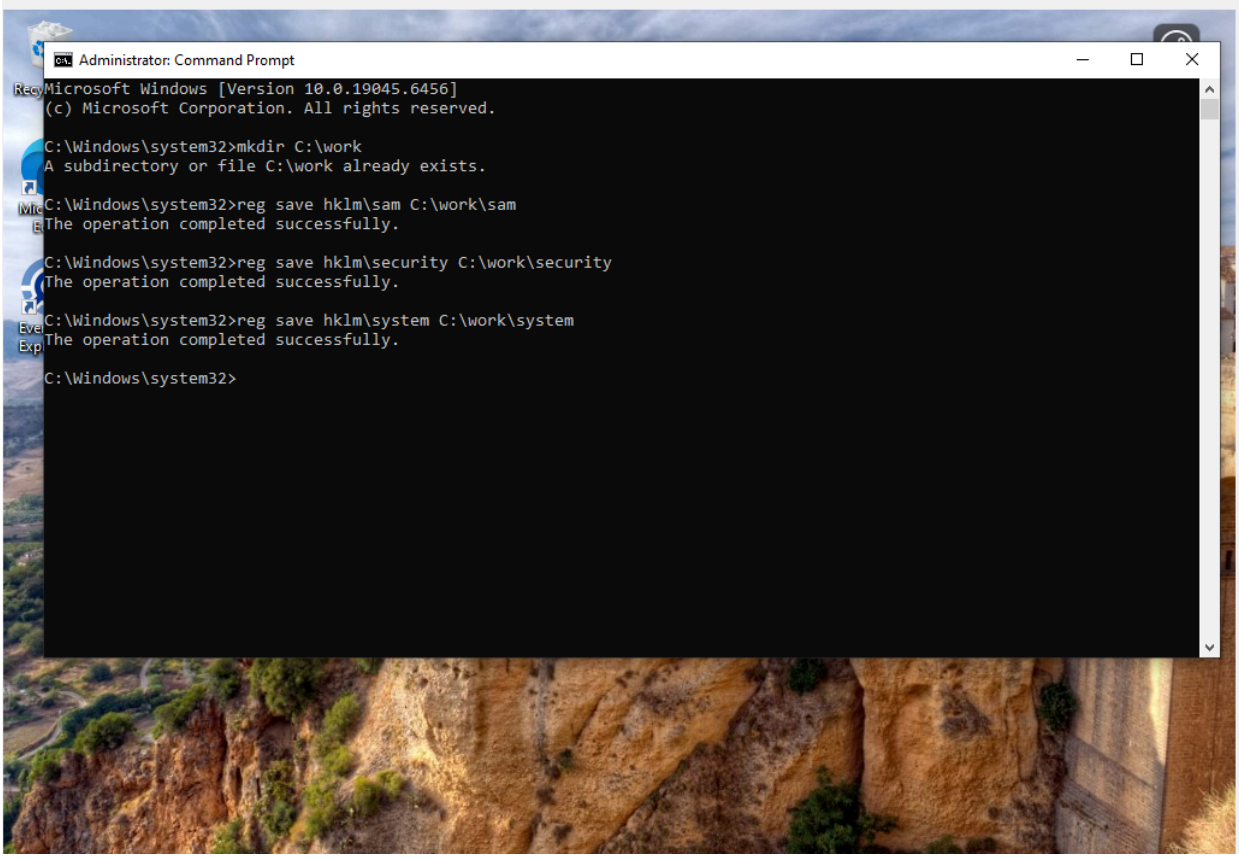

Part IV – Extracting Windows SAM Files

Next, I practiced extracting sensitive Windows registry hives that store system and user account data.
Using Command Prompt in Administrator mode, I saved the SAM, SECURITY, and SYSTEM registry hives into a forensic work folder (C:\work).

Commands executed:

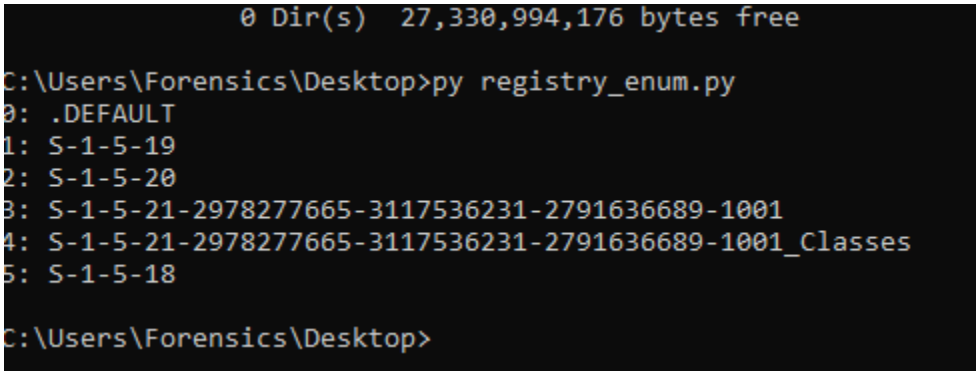
```
mkdir C:\work
reg save hklm\sam C:\work\sam
reg save hklm\security C:\work\security
reg save hklm\system C:\work\system
```

Each command returned “The operation completed successfully.”



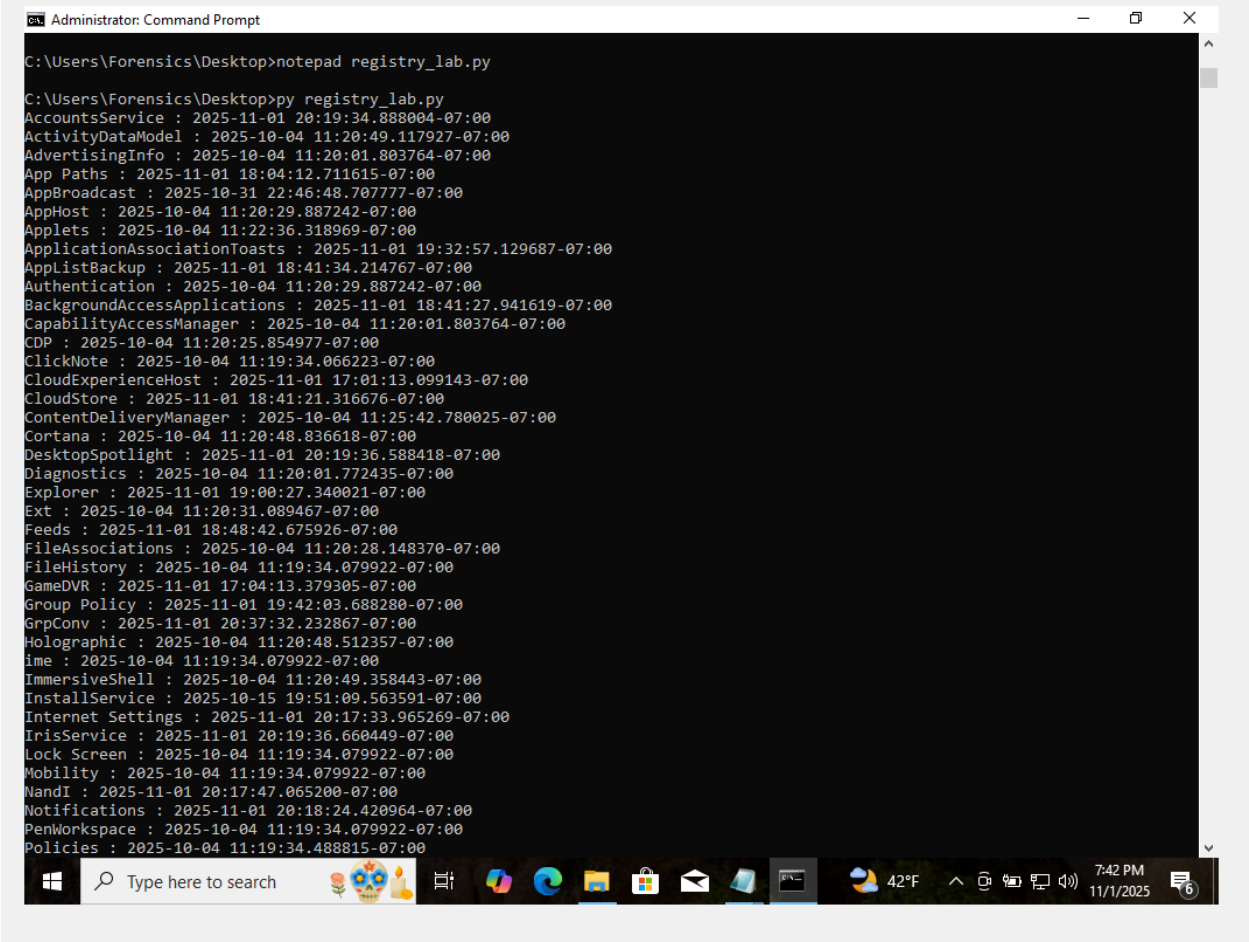

Part V – Handling Windows Registry using Python

In this section, I worked with Python's winreg library to enumerate registry keys and extract timestamps.
First, I listed the SIDs under HKEY_USERS, which identify each user account on the system.



)

Then, I accessed one SID's subkey at SOFTWARE\Microsoft\Windows\CurrentVersion to list and display the modification timestamps of its subkeys.



```
Administrator: Command Prompt

C:\Users\Forensics\Desktop>notepad registry_lab.py

C:\Users\Forensics\Desktop>py registry_lab.py
AccountsService : 2025-11-01 20:19:34.888004-07:00
ActivityDataModel : 2025-10-04 11:20:49.117927-07:00
AdvertisingInfo : 2025-10-04 11:20:01.803764-07:00
App Paths : 2025-11-01 18:04:12.711615-07:00
AppBroadcast : 2025-10-31 22:46:48.707777-07:00
AppHost : 2025-10-04 11:20:29.887242-07:00
Applets : 2025-10-04 11:22:36.318969-07:00
ApplicationAssociationToasts : 2025-11-01 19:32:57.129687-07:00
AppListBackup : 2025-11-01 18:41:34.214767-07:00
Authentication : 2025-10-04 11:20:29.887242-07:00
BackgroundAccessApplications : 2025-11-01 18:41:27.941619-07:00
CapabilityAccessManager : 2025-10-04 11:20:01.803764-07:00
CDP : 2025-10-04 11:20:25.854977-07:00
ClickNote : 2025-10-04 11:19:34.066223-07:00
CloudExperienceHost : 2025-11-01 17:01:13.099143-07:00
CloudStore : 2025-11-01 18:41:21.316676-07:00
ContentDeliveryManager : 2025-10-04 11:25:42.780025-07:00
Cortana : 2025-10-04 11:20:48.836618-07:00
DesktopSpotlight : 2025-11-01 20:19:36.588418-07:00
Diagnostics : 2025-10-04 11:20:01.772435-07:00
Explorer : 2025-11-01 19:00:27.340021-07:00
Ext : 2025-10-04 11:20:31.089467-07:00
Feeds : 2025-11-01 18:48:42.675926-07:00
FileAssociations : 2025-10-04 11:20:28.148370-07:00
FileHistory : 2025-10-04 11:19:34.079922-07:00
GameDVR : 2025-11-01 17:04:13.379305-07:00
Group Policy : 2025-11-01 19:42:03.688280-07:00
GrpConv : 2025-11-01 20:37:32.232867-07:00
Holographic : 2025-10-04 11:20:48.512357-07:00
ime : 2025-10-04 11:19:34.079922-07:00
ImmersiveShell : 2025-10-04 11:20:49.358443-07:00
InstallService : 2025-10-15 19:51:09.563591-07:00
Internet Settings : 2025-11-01 20:17:33.965269-07:00
IrisService : 2025-11-01 20:19:36.660449-07:00
Lock Screen : 2025-10-04 11:19:34.079922-07:00
Mobility : 2025-10-04 11:19:34.079922-07:00
NandI : 2025-11-01 20:17:47.065200-07:00
Notifications : 2025-11-01 20:18:24.420964-07:00
PenWorkspace : 2025-10-04 11:19:34.079922-07:00
Policies : 2025-10-04 11:19:34.488815-07:00
```

This helped demonstrate how investigators can determine when specific configuration changes occurred inside the registry.

Part VI – Windows Recycle Bin Analysis with Python

For this part, I used the winshell library to enumerate deleted files and recover metadata from the Recycle Bin. I also added pywin32 to ensure compatibility.
The script listed files that had been deleted, their recycle dates, and the associated user SID. I also created and deleted a test file on my Desktop to confirm the script worked.

```
C:\Users\Forensics\Desktop>notepad recycle_lab.py

C:\Users\Forensics\Desktop>py recycle_lab.py
C:\Users\Forensics\Desktop\test 2025-11-02 02:55:39+00:00
C:\Users\Forensics\Desktop\New folder   2025-10-04 18:26:03+00:00
S-1-5-21-2978277665-3117536231-2791636689-1001
After delete:
C:\Users\Forensics\Desktop\test 2025-11-02 02:55:39+00:00
C:\Users\Forensics\Desktop\test 2025-11-02 02:59:22+00:00
C:\Users\Forensics\Desktop\New folder   2025-10-04 18:26:03+00:00

C:\Users\Forensics\Desktop>
```

This test confirmed how deleted items can still be located, time-stamped, and tied to a specific user account.

Part VII – Reading Browser History using Python

Finally, I used Python with sqlite3 to extract history data from Firefox's places.sqlite database. The script located the profile folder, listed all .sqlite files, and queried URLs with visit counts.

```
:\Users\Forensics\Desktop>py firefox_history.py
rofiles: ['mjfk6ml0.default-release', 'nt02vb35.default']
sing profile: C:\Users\Forensics\AppData\Roaming\Mozilla\Firefox\Profiles\mjfk6ml0.default-release
ounce-tracking-protection.sqlite
ert9.db
ontent-prefs.sqlite
ookies.sqlite
omain_to_categories.sqlite
avicons.sqlite
ormhistory.sqlite
ey4.db
ermissions.sqlite
laces.sqlite
rotections.sqlite
torage.sqlite
uggest.sqlite
ebappsstore.sqlite
'https://support.mozilla.org/products/firefox', 0)
'https://support.mozilla.org/kb/customize-firefox-controls-buttons-and-toolbars?utm_source=firefox-browser&utm_medium=defaul
-bookmarks&utm_campaign=customize', 0)
'https://www.mozilla.org/contribute/', 0)
'https://www.mozilla.org/about/', 0)
'https://www.wikipedia.org/', 1)

:\Users\Forensics\Desktop>
```

This provided evidence of recently visited websites and the number of times each was accessed, which is crucial in timeline reconstruction.

Conclusion

Through this lab, I was able to analyze key artifacts that reveal user activity in a Windows environment. I learned how to extract browser data, interpret event logs, and query registry hives programmatically. These techniques demonstrate how forensic investigators recover and interpret digital evidence to trace user behavior, system events, and data remnants left behind even after deletion.